

# Database System Engineering and Distributed Backend Development Course

Code: 25CS1302E

ROLL NO : 2520030255

NAME : V SAI NIKHIL

## WEEK-4 MYSQL VIEWS

```
1 • CREATE DATABASE IF NOT EXISTS employeeedb1;
2 • USE employeeedb1;
3
4 • CREATE TABLE IF NOT EXISTS departments (
5     dept_id INT PRIMARY KEY,
6     dept_name VARCHAR(50)
7 );
8
9 • CREATE TABLE IF NOT EXISTS employees (
10     emp_id INT PRIMARY KEY,
11     emp_name VARCHAR(50),
12     dept_id INT,
13     salary DECIMAL(10,2)
14 );
15
16 • INSERT IGNORE INTO departments VALUES
17     (1,'CSE'),
18     (2,'ECE'),
19     (3,'EEE');
20
21 • INSERT IGNORE INTO employees VALUES
22     (101,'Rahul',1,50000),
23     (102,'Priya',2,65000),
24     (103,'Anil',1,55000),
25     (104,'Sneha',3,70000),
```

Output

Action Output

| #     | Time     | Action                                                                                                        | Message                                                |
|-------|----------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| ✓ 96  | 13:12:20 | CREATE DATABASE IF NOT EXISTS employeeedb1                                                                    | 1 row(s) affected                                      |
| ✓ 97  | 13:12:20 | USE employeeedb1                                                                                              | 0 row(s) affected                                      |
| ✓ 98  | 13:12:20 | CREATE TABLE IF NOT EXISTS departments ( dept_id INT PRIMARY KEY, dept_name VARCHAR(50) )                     | 0 row(s) affected                                      |
| ✓ 99  | 13:12:20 | CREATE TABLE IF NOT EXISTS employees ( emp_id INT PRIMARY KEY, emp_name VARCHAR(50), d...                     | 0 row(s) affected                                      |
| ✓ 100 | 13:12:20 | INSERT IGNORE INTO departments VALUES (1,'CSE'), (2,'ECE'), (3,'EEE')                                         | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 |
| ✓ 101 | 13:12:20 | INSERT IGNORE INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), (1... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 |

```
1 SHOW TABLES;
2 • SELECT * FROM departments;
3 • SELECT * FROM employees;
4
```

| Result Grid |        |          |              |          |
|-------------|--------|----------|--------------|----------|
|             |        |          | Filter Rows: |          |
|             | emp_id | emp_name | dept_id      | salary   |
| ▶           | 101    | Rahul    | 1            | 50000.00 |
|             | 102    | Priya    | 2            | 65000.00 |
|             | 103    | Anil     | 1            | 55000.00 |
|             | 104    | Sneha    | 3            | 70000.00 |
|             | 105    | Kiran    | 2            | 48000.00 |
| •           | NULL   | NULL     | NULL         | NULL     |

```

1 DROP VIEW IF EXISTS EmployeeView;
2
3 • CREATE VIEW EmployeeView AS
4 SELECT *
5 FROM employees;
6
7 • SELECT * FROM EmployeeView;
8

```

| Result Grid |        |              |         |                                                                                                 |
|-------------|--------|--------------|---------|-------------------------------------------------------------------------------------------------|
|             |        | Filter Rows: |         | Export:    W |
|             | emp_id | emp_name     | dept_id | salary                                                                                          |
| ▶           | 101    | Rahul        | 1       | 50000.00                                                                                        |
|             | 102    | Priya        | 2       | 65000.00                                                                                        |
|             | 103    | Anil         | 1       | 55000.00                                                                                        |
|             | 104    | Sneha        | 3       | 70000.00                                                                                        |
|             | 105    | Kiran        | 2       | 48000.00                                                                                        |

```

1 DROP VIEW IF EXISTS EmployeeView;
2
3 • CREATE VIEW EmployeeView AS
4 SELECT *
5 FROM employees;
6
7 • SELECT * FROM EmployeeView;
8

```

| Result Grid |        |              |         |                                                                                                 |
|-------------|--------|--------------|---------|-------------------------------------------------------------------------------------------------|
|             |        | Filter Rows: |         | Export:    W |
|             | emp_id | emp_name     | dept_id | salary                                                                                          |
| ▶           | 101    | Rahul        | 1       | 50000.00                                                                                        |
|             | 102    | Priya        | 2       | 65000.00                                                                                        |
|             | 103    | Anil         | 1       | 55000.00                                                                                        |
|             | 104    | Sneha        | 3       | 70000.00                                                                                        |
|             | 105    | Kiran        | 2       | 48000.00                                                                                        |

```
1 • DROP VIEW IF EXISTS EmployeeBasicView;
2
3 • CREATE VIEW EmployeeBasicView AS
4   SELECT emp_id, emp_name, salary
5   FROM employees;
6
7 • SELECT * FROM EmployeeBasicView;
8
```

Result Grid



Filter Rows:

Export:



|   | emp_id | emp_name | salary   |
|---|--------|----------|----------|
| ▶ | 101    | Rahul    | 50000.00 |
|   | 102    | Priya    | 65000.00 |
|   | 103    | Anil     | 55000.00 |
|   | 104    | Sneha    | 70000.00 |
|   | 105    | Kiran    | 48000.00 |

```
1 • DROP VIEW IF EXISTS HighSalaryEmployees;
2
3 CREATE VIEW HighSalaryEmployees AS
4 SELECT emp_id, emp_name, salary
5 FROM employees
6 WHERE salary > 55000;
7
8 • SELECT * FROM HighSalaryEmployees;
9
```

| Result Grid                                   |        |          |          |
|-----------------------------------------------|--------|----------|----------|
| Filter Rows: <input type="text"/>             |        |          |          |
| Export: <input type="button" value="Export"/> |        |          |          |
| Wrap                                          |        |          |          |
|                                               | emp_id | emp_name | salary   |
| ▶                                             | 102    | Priya    | 65000.00 |
|                                               | 104    | Sneha    | 70000.00 |

```

3 • CREATE VIEW EmployeeDepartmentView AS
4 SELECT
5     e.emp_id,
6     e.emp_name,
7     d.dept_name,
8     e.salary
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id;
12
13 • SELECT * FROM EmployeeDepartmentView;
14

```

Result Grid



Filter Rows:

Export:



Wra

|   | emp_id | emp_name | dept_name | salary   |
|---|--------|----------|-----------|----------|
| ▶ | 101    | Rahul    | CSE       | 50000.00 |
|   | 102    | Priya    | ECE       | 65000.00 |
|   | 103    | Anil     | CSE       | 55000.00 |
|   | 104    | Sneha    | EEE       | 70000.00 |
|   | 105    | Kiran    | ECE       | 48000.00 |

```
4      SELECT
5          e.emp_id,
6          e.emp_name,
7          d.dept_name,
8          e.salary
9      FROM employees e
10     JOIN departments d
11     ON e.dept_id = d.dept_id
12     WHERE d.dept_name = 'CSE';
13
14 •    SELECT * FROM CSEEmployees;
15
```

Result Grid   Filter Rows:

|   | emp_id | emp_name | dept_name | salary   |
|---|--------|----------|-----------|----------|
| ▶ | 101    | Rahul    | CSE       | 50000.00 |
|   | 103    | Anil     | CSE       | 55000.00 |

```

4 • CREATE VIEW DepartmentSalaryView AS
5 SELECT
6     d.dept_name,
7     COUNT(e.emp_id) AS EmployeeCount,
8     AVG(e.salary) AS AverageSalary
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id
12 GROUP BY d.dept_id, d.dept_name;
13
14 • SELECT * FROM DepartmentSalaryView;
15

```

Result Grid



Filter Rows:

Export:



Wrap C

|   | dept_name | EmployeeCount | AverageSalary |
|---|-----------|---------------|---------------|
| ► | CSE       | 2             | 52500.000000  |
|   | ECE       | 2             | 56500.000000  |
|   | EEE       | 1             | 70000.000000  |

```

3 • CREATE VIEW HighAverageDepartments AS
4 SELECT
5     d.dept_name,
6     AVG(e.salary) AS AverageSalary
7 FROM employees e
8 JOIN departments d
9 ON e.dept_id = d.dept_id
10 GROUP BY d.dept_id, d.dept_name
11 HAVING AVG(e.salary) > 55000;
12
13 • SELECT * FROM HighAverageDepartments;
14

```

Result Grid |   Filter Rows:  | Export:  | Wrap C

| dept_name | AverageSalary |
|-----------|---------------|
| ECE       | 56500.000000  |
| EEE       | 70000.000000  |

```
1
2 • DROP VIEW IF EXISTS SalaryRangeView;
3
4 • CREATE VIEW SalaryRangeView AS
5   SELECT emp_id, emp_name, salary
6   FROM employees
7   WHERE salary BETWEEN 50000 AND 70000;
8
9 • SELECT * FROM SalaryRangeView;
10
11
```

| Result Grid                                                                                                                                                                                           |        |          |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------|----------|
|   Filter Rows: <input type="text"/> |        |          |          |
| Export:  Wrap                                                                                                      |        |          |          |
|                                                                                                                                                                                                       | emp_id | emp_name | salary   |
| ▶                                                                                                                                                                                                     | 101    | Rahul    | 50000.00 |
|                                                                                                                                                                                                       | 102    | Priya    | 65000.00 |
|                                                                                                                                                                                                       | 103    | Anil     | 55000.00 |
|                                                                                                                                                                                                       | 104    | Sneha    | 70000.00 |

```

1 • DROP VIEW IF EXISTS AnnualSalaryView;
2
3 • CREATE VIEW AnnualSalaryView AS
4     SELECT
5         emp_id,
6         emp_name,
7         salary AS MonthlySalary,
8         salary * 12 AS AnnualSalary
9     FROM employees;
10
11 • SELECT * FROM AnnualSalaryView;
12

```

Result Grid |   Filter Rows:  | Export: 

|   | emp_id | emp_name | MonthlySalary | AnnualSalary |
|---|--------|----------|---------------|--------------|
| ▶ | 101    | Rahul    | 50000.00      | 600000.00    |
|   | 102    | Priya    | 65000.00      | 780000.00    |
|   | 103    | Anil     | 55000.00      | 660000.00    |
|   | 104    | Sneha    | 70000.00      | 840000.00    |
|   | 105    | Kiran    | 48000.00      | 576000.00    |

```
2
3 • CREATE VIEW EmployeeSalaryView AS
4   SELECT emp_id, emp_name, salary
5   FROM employees;
6
7 • UPDATE EmployeeSalaryView
8   SET salary = 60000
9   WHERE emp_id = 101;
10
11 • SELECT * FROM EmployeeSalaryView;
12 • SELECT * FROM employees WHERE emp_id = 101;
13
```

|             |                                                                                   |                                                                                   |                                   |                                                                                           |                                                                                     |                                                                                     |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Result Grid |  |  | Filter Rows: <input type="text"/> | Edit:  |  |  |
|             | emp_id                                                                            | emp_name                                                                          | dept_id                           | salary                                                                                    |                                                                                     |                                                                                     |
| ▶           | 101                                                                               | Rahul                                                                             | 1                                 | 60000.00                                                                                  |                                                                                     |                                                                                     |
| *           | NULL                                                                              | NULL                                                                              | NULL                              | NULL                                                                                      |                                                                                     |                                                                                     |

# Database System Engineering and Distributed Backend Development Course

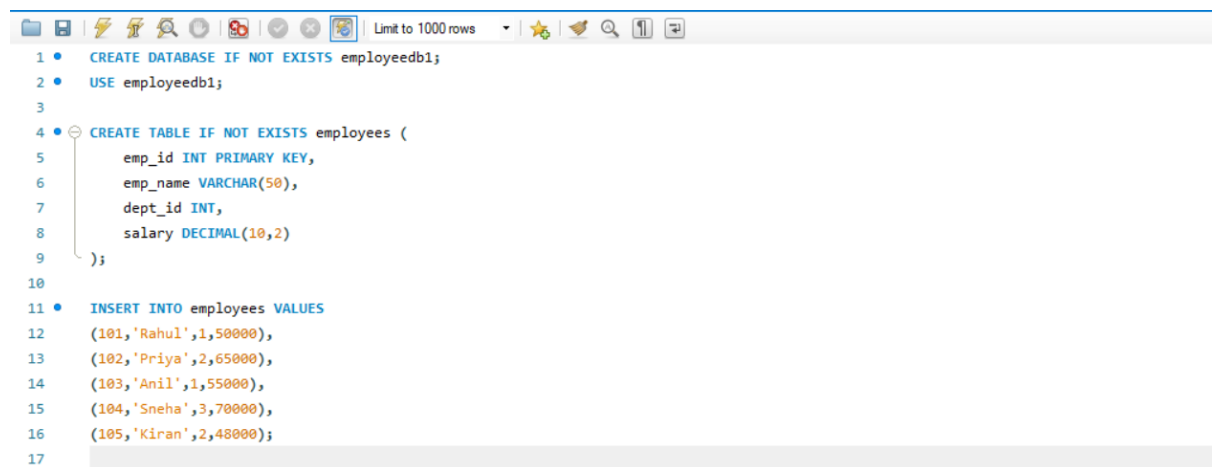
Code: 25CS1302E

ROLL NO : 2520030255

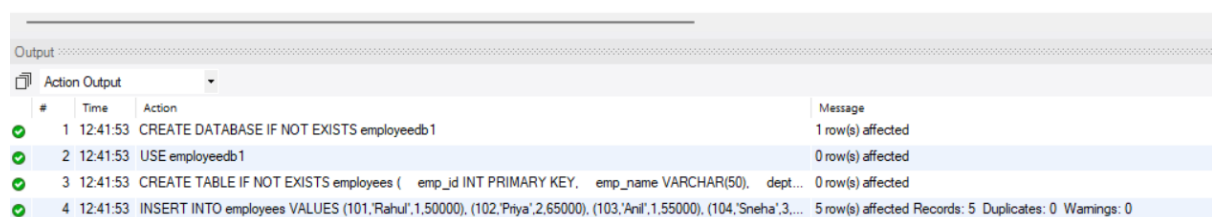
NAME : V SAI NIKHIL

## WEEK-4 MYSQL STORED PROCEDURES

Setup: employee table



```
1 • CREATE DATABASE IF NOT EXISTS employeeeb1;
2 • USE employeeeb1;
3
4 • CREATE TABLE IF NOT EXISTS employees (
5     emp_id INT PRIMARY KEY,
6     emp_name VARCHAR(50),
7     dept_id INT,
8     salary DECIMAL(10,2)
9 );
10
11 • INSERT INTO employees VALUES
12     (101,'Rahul',1,50000),
13     (102,'Priya',2,65000),
14     (103,'Anil',1,55000),
15     (104,'Sneha',3,70000),
16     (105,'Kiran',2,48000);
17
```



| #   | Time     | Action                                                                                                              | Message                                                |
|-----|----------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| ✓ 1 | 12:41:53 | CREATE DATABASE IF NOT EXISTS employeeeb1                                                                           | 1 row(s) affected                                      |
| ✓ 2 | 12:41:53 | USE employeeeb1                                                                                                     | 0 row(s) affected                                      |
| ✓ 3 | 12:41:53 | CREATE TABLE IF NOT EXISTS employees ( emp_id INT PRIMARY KEY, emp_name VARCHAR(50), dept...                        | 0 row(s) affected                                      |
| ✓ 4 | 12:41:53 | INSERT INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), (104,'Sneha',3,... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 |

verify data

```
1 • SELECT * FROM employees;
```

Result Grid




Filter Rows:

Edit:





Ex

|   | emp_id | emp_name | dept_id | salary   |
|---|--------|----------|---------|----------|
| ▶ | 101    | Rahul    | 1       | 50000.00 |
|   | 102    | Priya    | 2       | 65000.00 |
|   | 103    | Anil     | 1       | 55000.00 |
|   | 104    | Sneha    | 3       | 70000.00 |
|   | 105    | Kiran    | 2       | 48000.00 |
| ✱ | NULL   | NULL     | NULL    | NULL     |

### Display all employees

```
1      DROP PROCEDURE IF EXISTS GetEmployees;
2      DELIMITER $$
3
4  •   CREATE PROCEDURE GetEmployees()
5      BEGIN
6          SELECT * FROM employees;
7      END $$
8
9      DELIMITER ;
10
11 •   CALL GetEmployees();
12
```

| Result Grid |        |          |         |          | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|--------|----------|---------|----------|--------------|---------|--------------------|
|             | emp_id | emp_name | dept_id | salary   |              |         |                    |
| ▶           | 101    | Rahul    | 1       | 50000.00 |              |         |                    |
|             | 102    | Priya    | 2       | 65000.00 |              |         |                    |
|             | 103    | Anil     | 1       | 55000.00 |              |         |                    |
|             | 104    | Sneha    | 3       | 70000.00 |              |         |                    |
|             | 105    | Kiran    | 2       | 48000.00 |              |         |                    |

in parametre

```
1 • DROP PROCEDURE IF EXISTS GetEmployeeById;
2 DELIMITER $$
3
4 • CREATE PROCEDURE GetEmployeeById(IN eid INT)
5 BEGIN
6     SELECT * FROM employees
7     WHERE emp_id = eid;
8 END $$
9
10 DELIMITER ;
11
12 • CALL GetEmployeeById(102);
13
14
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



|   | emp_id | emp_name | dept_id | salary   |
|---|--------|----------|---------|----------|
| ► | 102    | Priya    | 2       | 65000.00 |

## update salary

```
4
5 • CREATE PROCEDURE IncreaseSalary(
6     IN eid INT,
7     IN amount DECIMAL(10,2)
8 )
9 BEGIN
10     UPDATE employees
11     SET salary = salary + amount
12     WHERE emp_id = eid;
13 END $$
14
15 DELIMITER ;
16
17 • CALL IncreaseSalary(101,5000);
18 • SELECT * FROM employees WHERE emp_id=101;
19
```

Result Grid

Filter Rows:

Edit:    Export

|   | emp_id | emp_name | dept_id | salary   |
|---|--------|----------|---------|----------|
| ▶ | 101    | Rahul    | 1       | 55000.00 |
| * | NULL   | NULL     | NULL    | NULL     |

## join

```

2 • DROP PROCEDURE IF EXISTS AvgSalary;
3 DELIMITER $$
4
5 • CREATE PROCEDURE AvgSalary()
6 BEGIN
7     SELECT AVG(salary) AS AverageSalary
8     FROM employees;
9 END $$
10
11 DELIMITER ;
12
13 • CALL AvgSalary();
14

```

| Result Grid |               | Filter Rows: | Export: | Wrap Cell Content |
|-------------|---------------|--------------|---------|-------------------|
|             | AverageSalary |              |         |                   |
| ▶           | 58600.000000  |              |         |                   |

## Group by and avg

```
8
9 • DROP PROCEDURE IF EXISTS EmployeeDepartment;
10 DELIMITER $$
11
12 • CREATE PROCEDURE EmployeeDepartment()
13 BEGIN
14     SELECT e.emp_name, d.dept_name, e.salary
15     FROM employees e
16     JOIN departments d ON e.dept_id=d.dept_id;
17 END $$
18
19 DELIMITER ;
20
21 • CALL EmployeeDepartment();
22
```

Result Grid |  Filter Rows:  | Export:  | Wrap Cell Content: 

|   | emp_name | dept_name | salary   |
|---|----------|-----------|----------|
| ▶ | Rahul    | CSE       | 55000.00 |
|   | Priya    | ECE       | 65000.00 |
|   | Anil     | CSE       | 55000.00 |
|   | Sneha    | EEE       | 70000.00 |
|   | Kiran    | ECE       | 48000.00 |

If else :

```
4 • CREATE PROCEDURE DepartmentAverageSalary()
5 BEGIN
6     SELECT d.dept_name,
7           COUNT(e.emp_id) AS EmployeeCount,
8           AVG(e.salary) AS AverageSalary
9     FROM employees e
10    JOIN departments d ON e.dept_id=d.dept_id
11   GROUP BY d.dept_id, d.dept_name;
12 END $$
13
14 DELIMITER ;
15
16 • CALL DepartmentAverageSalary();
17
```

| Result Grid  |           |               |                    |
|--------------|-----------|---------------|--------------------|
| Filter Rows: |           | Export:       | Wrap Cell Content: |
|              | dept_name | EmployeeCount | AverageSalary      |
| ▶            | CSE       | 2             | 55000.000000       |
|              | ECE       | 2             | 56500.000000       |
|              | EEE       | 1             | 70000.000000       |

```

8      SELECT salary INTO sal
9      FROM employees
10     WHERE emp_id=eid;
11
12     IF sal >= 60000 THEN
13         SELECT 'High Salary' AS Status;
14     ELSE
15         SELECT 'Low Salary' AS Status;
16     END IF;
17 END $$
18
19 DELIMITER ;
20
21 • CALL SalaryStatus(102);
22

```

| Result Grid |             | Filter Rows: | Export: | Wrap Cell Co |
|-------------|-------------|--------------|---------|--------------|
|             | Status      |              |         |              |
| ▶           | High Salary |              |         |              |

Out parameter

### Inout parameter

```
1 • DROP PROCEDURE IF EXISTS EmployeeCount;
2 DELIMITER $$
3
4 • CREATE PROCEDURE EmployeeCount(OUT total INT)
5 BEGIN
6     SELECT COUNT(*) INTO total
7     FROM employees;
8 END $$
9
10 DELIMITER ;
11
12 • CALL EmployeeCount(@total);
13 • SELECT @total AS TotalEmployees;
14
15
```

| Result Grid |                | Filter Rows: | Export: | Wrap Cell Cont |
|-------------|----------------|--------------|---------|----------------|
|             | TotalEmployees |              |         |                |
| ▶           | 5              |              |         |                |

transaction

```
1 • DROP PROCEDURE IF EXISTS AddBonus;
2 DELIMITER $$
3
4 • CREATE PROCEDURE AddBonus(INOUT bonus DECIMAL(10,2))
5 BEGIN
6     SET bonus = bonus + 5000;
7 END $$
8
9 DELIMITER ;
10
11 • SET @bonus=10000;
12 • CALL AddBonus(@bonus);
13 • SELECT @bonus AS UpdatedBonus;
14
15
```

Result Grid   Filter Rows:  Export:  Wrap Cell Content: 

|   | UpdatedBonus |
|---|--------------|
| ▶ | 15000.00     |

```

13      SET salary=salary-amount
14      WHERE emp_id=fromEmp;
15
16      UPDATE employees
17      SET salary=salary+amount
18      WHERE emp_id=toEmp;
19
20      COMMIT;
21  END $$
22
23  DELIMITER ;
24
25 • CALL TransferSalaryAmount(101,102,2000);
26 • SELECT * FROM employees WHERE emp_id IN (101,102);
27

```

| Result Grid | Filter Rows: | Edit:   | Export/Import: |
|-------------|--------------|---------|----------------|
| emp_id      | emp_name     | dept_id | salary         |
| 101         | Rahul        | 1       | 53000.00       |
| 102         | Priya        | 2       | 67000.00       |
| NULL        | NULL         | NULL    | NULL           |

```

1 • SHOW PROCEDURE STATUS
2 WHERE Db='employeeeb1';
3
4 SHOW CREATE PROCEDURE GetEmployees;
5
6 • DROP PROCEDURE IF EXISTS GetEmployees;
7

```

|             |              |                                          |                                            |                      |                      |                    |  |              |                      |  |         |  |                    |                      |
|-------------|--------------|------------------------------------------|--------------------------------------------|----------------------|----------------------|--------------------|--|--------------|----------------------|--|---------|--|--------------------|----------------------|
| Result Grid |              |                                          |                                            |                      |                      |                    |  | Filter Rows: | <input type="text"/> |  | Export: |  | Wrap Cell Content: | <input type="text"/> |
|             | Procedure    | sql_mode                                 | Create Procedure                           | character_set_client | collation_connection | Database Collation |  |              |                      |  |         |  |                    |                      |
| ▶           | GetEmployees | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE... | CREATE DEFINER='root'@'localhost' PROCE... | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4_0900_ai_ci |  |              |                      |  |         |  |                    |                      |

